

HW_04

Henry Traynor

Due: 5:00pm, 15 September

Create a self-contained R script or Rmarkdown/Quarto file to do the following:

Using the simulated data, compare the prior and posterior predictive distributions of λ for each of the 3 forest types.

Hint: because forest type is categorical (not continuous), you can visualize the prior and posterior predictive distributions as histograms for each forest type.

```
set.seed(314159) # for reproducibility

nSites <- 100
nVisits <- 4

forest <- factor(sample(c("Hardwood", "Mixed", "Pine"), nSites, replace=TRUE))
forestMixed <- ifelse(forest=="Mixed", 1, 0) ## Dummy
forestPine <- ifelse(forest=="Pine", 1, 0) ## Dummy
temp <- matrix(rnorm(nSites*nVisits), nrow=nSites, ncol=nVisits)

# true coeffs
beta0 <- 0
beta1 <- -1
beta2 <- 1
lambda <- exp(beta0 + beta1*forestMixed + beta2*forestPine)
alpha0 <- -2
alpha1 <- 1
p <- plogis(alpha0 + alpha1*temp)

# simulate occupancy and detection data
N <- rpois(nSites, lambda=lambda) ## local abundance
y <- matrix(NA, nrow=nSites, ncol=nVisits)
for(i in 1:nSites) {
  y[i,] <- rbinom(nVisits, size=N[i], prob=p[i,])
}

# summary statistics
maxCounts <- apply(y, 1, max)
table(maxCounts)

## maxCounts
## 0 1 2 3 4
## 61 34 2 2 1
```

```
naiveOccupancy <- sum(maxCounts>0)/nSites
naiveOccupancy
```

```
## [1] 0.39
```

```
naiveAbund <- sum(maxCounts)
naiveAbund
```

```
## [1] 48
```

```
# prep data in named list
jags.data <- list(y=y,
                 forestMixed=forestMixed,
                 forestPine=forestPine,
                 nSites=nSites,
                 temp=temp,
                 nOccasions=nVisits)
# initial values for MCMC
jags.inits <- function() {
  list(beta0=rnorm(1), alpha0=rnorm(1), N=maxCounts)
}
# parameters to save
jags.pars <- c("beta0", "beta1", "beta2",
              "alpha0", "alpha1", "totalAbundance")

# run model
library(jagsUI)
jags.post.samples <- jags.basic(data=jags.data,
                                inits=jags.inits,
                                parameters.to.save=c(jags.pars,"N"),
                                model.file="Nmix-model-covs.jag",
                                n.chains=3,
                                n.adapt=100,
                                n.burnin=0,
                                n.iter=2000,
                                parallel=TRUE)
```

```
##
## Processing function input.....
##
## Done.
##
## Beginning parallel processing using 3 cores. Console output will be suppressed.
##
## Parallel processing completed.
##
## MCMC took 0.047 minutes.
```

```
summary(jags.post.samples[,jags.pars])
```

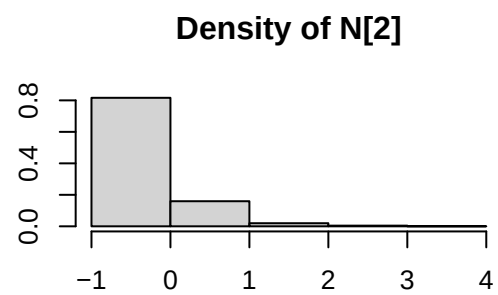
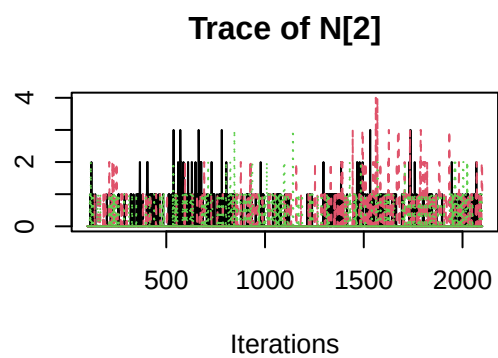
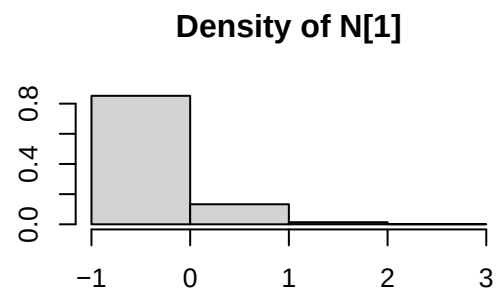
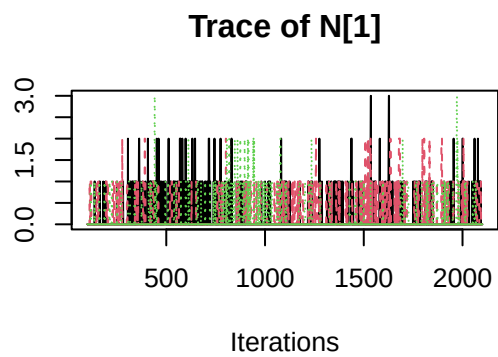
```
##
```

```

## Iterations = 101:2100
## Thinning interval = 1
## Number of chains = 3
## Sample size per chain = 2000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##              Mean      SD Naive SE Time-series SE
## beta0        -0.2328  0.2851 0.003681      0.03369
## beta1        -0.7992  0.4155 0.005364      0.04327
## beta2         0.8156  0.3186 0.004112      0.02524
## alpha0       -1.5684  0.3836 0.004952      0.10263
## alpha1         1.0592  0.1637 0.002113      0.02087
## totalAbundance 92.4773 22.6984 0.293035      5.82202
##
## 2. Quantiles for each variable:
##
##              2.5%    25%    50%    75%    97.5%
## beta0        -0.7760 -0.4277 -0.2497 -0.04262  0.366734
## beta1        -1.6647 -1.0528 -0.7880 -0.52895 -0.001139
## beta2         0.2006  0.6004  0.8134  1.02425  1.463198
## alpha0       -2.2776 -1.8919 -1.5084 -1.29019 -0.895715
## alpha1         0.7402  0.9532  1.0498  1.16248  1.397912
## totalAbundance 62.0000 76.0000 86.0000 105.00000 146.000000

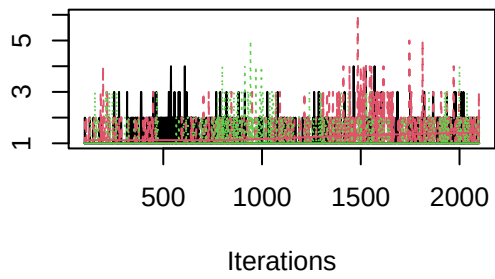
```

```
plot(jags.post.samples[,paste0("N[", 1:2, "]")])
```

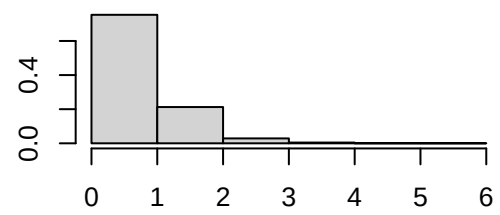


```
plot(jags.post.samples[,paste0("N[", 3:4, "]")])
```

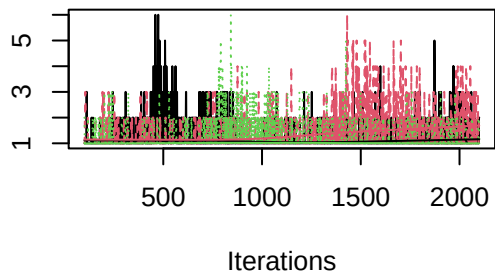
Trace of N[3]



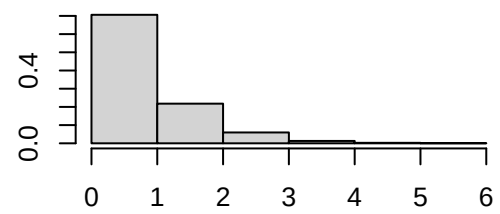
Density of N[3]



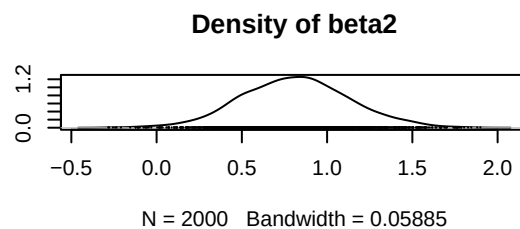
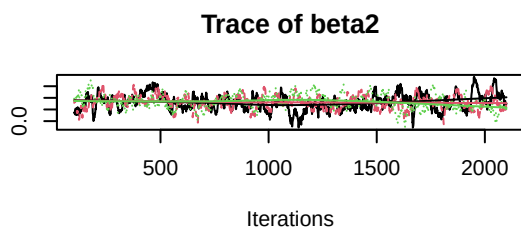
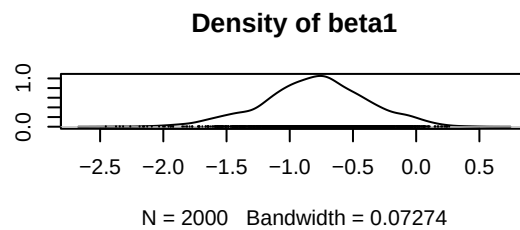
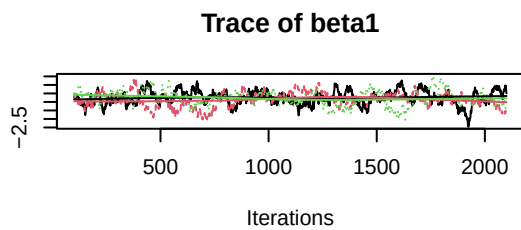
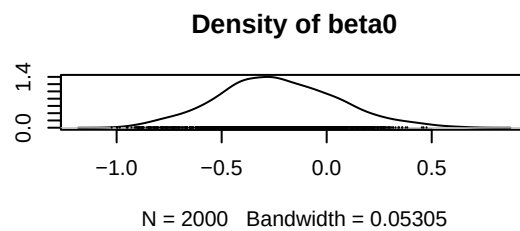
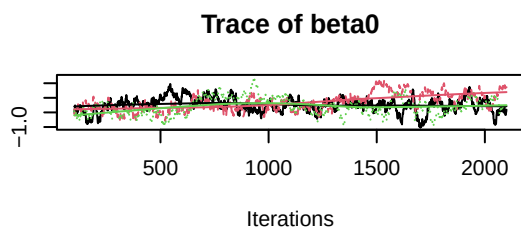
Trace of N[4]



Density of N[4]

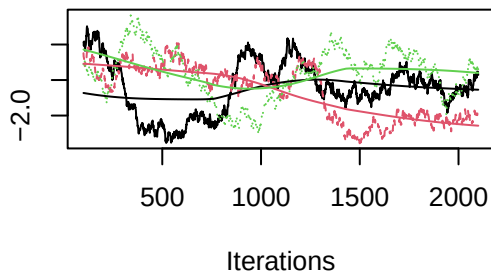


```
plot(jags.post.samples[,jags.pars[1:3]])
```

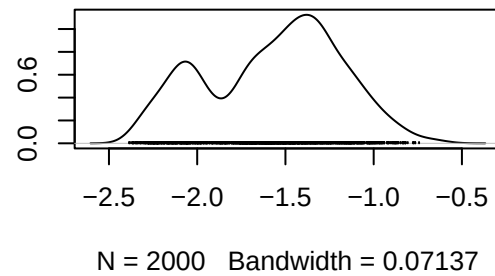


```
plot(jags.post.samples[,jags.pars[4:5]])
```

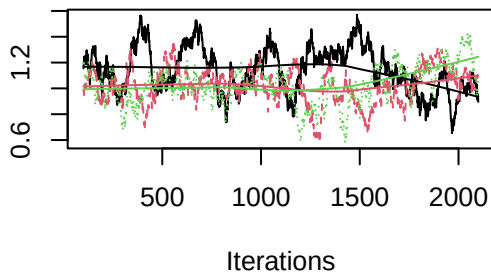
Trace of alpha0



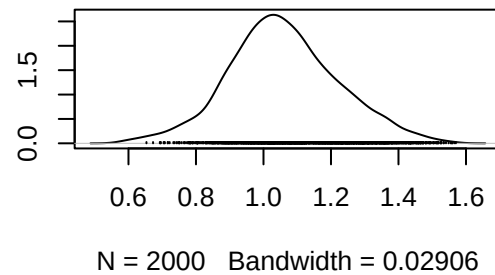
Density of alpha0



Trace of alpha1



Density of alpha1



```
lambda.coef.post <- as.matrix(jags.post.samples[,c("beta0", "beta1", "beta2")])
head(lambda.coef.post, n=4)
```

```
##           beta0      beta1      beta2
## [1,] -0.4837185 -0.7557389 0.3601422
## [2,] -0.4710857 -0.6743193 0.3341998
## [3,] -0.3907293 -0.7037765 0.3400097
## [4,] -0.3746024 -0.7438384 0.4104394
```

```
n.iter <- nrow(lambda.coef.post)
forest.pred <- c(rep("Hardwood", 50),
                 rep("Mixed", 50),
                 rep("Pine", 50))
```

```
forest.Mixed.pred <- ifelse(forest.pred=="Mixed", 1, 0)
forest.Pine.pred <- ifelse(forest.pred=="Pine", 1, 0)
```

```
lambda.post.pred <- matrix(NA, nrow=n.iter, ncol=length(forest.pred))
```

```
for(i in 1:n.iter) {
  lambda.post.pred[i,] <- plogis(lambda.coef.post[i,"beta0"] + lambda.coef.post[i,"beta1"]*forest.Mixed.pred[i] + lambda.coef.post[i,"beta2"]*forest.Pine.pred[i])
}
```

plot showing forest type on categorical horiz. axis and lambda on continuous vert. vertical axis

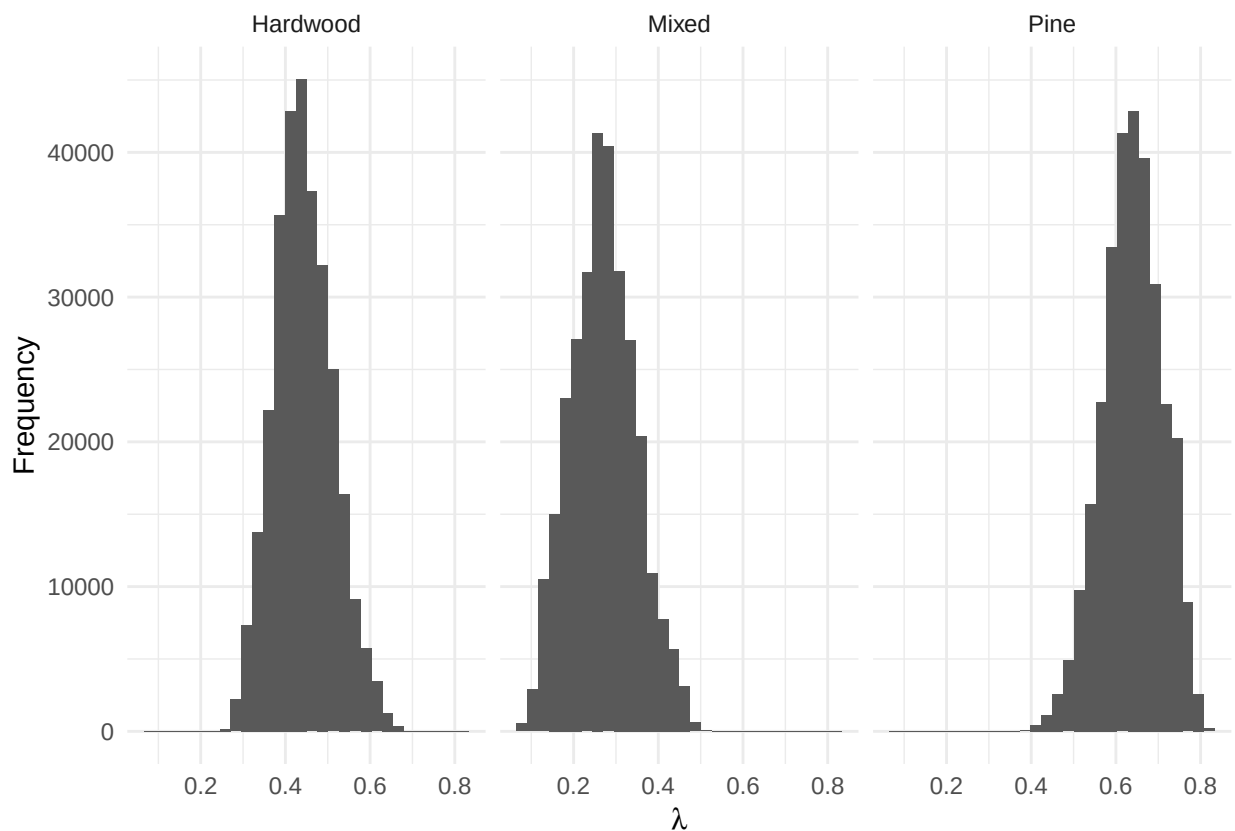
```

hardwood <- as.vector(lambda.post.pred[, 1:50])
mixed    <- as.vector(lambda.post.pred[, 51:100])
pine     <- as.vector(lambda.post.pred[, 101:150])

lambda.long <- data.frame(
  lambda = c(hardwood, mixed, pine),
  forest = rep(
    c("Hardwood", "Mixed", "Pine"),
    times = c(length(hardwood), length(mixed), length(pine))
  )
)

library(ggplot2)
ggplot(lambda.long, aes(x = lambda)) +
  geom_histogram(bins = 30) +
  facet_wrap(~forest, ncol = 3) +
  labs(
    x = expression(lambda),
    y = "Frequency"
  ) +
  theme_minimal()

```



```

jags.data2 <- jags.data
jags.data2$y[] <- NA      # Replace data with missing values

```

```
jags.prior.samples <- jags.basic(data=jags.data2,
                                inits=jags.inits,
                                parameters.to.save=jags.pars,
                                model.file="Nmix-model-covs.jag",
                                n.chains=3,
                                n.adapt=100,
                                n.burnin=0,
                                n.iter=2000,
                                parallel=TRUE)
```

```
##
## Processing function input.....
##
## Done.
##
## Beginning parallel processing using 3 cores. Console output will be suppressed.
##
## Parallel processing completed.
##
## MCMC took 0.009 minutes.
```

```
summary(jags.prior.samples)
```

```
##
## Iterations = 1:2000
## Thinning interval = 1
## Number of chains = 3
## Sample size per chain = 2000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##
```

	Mean	SD	Naive SE	Time-series SE
alpha0	-0.025196	1.410	0.01820	0.01787
alpha1	-0.004061	1.414	0.01825	0.01831
beta0	-0.026819	1.408	0.01818	0.01813
beta1	-0.003649	1.446	0.01867	0.01867
beta2	-0.016077	1.405	0.01814	0.01793
totalAbundance	571.981167	3195.146	41.24916	41.25492

```
##
## 2. Quantiles for each variable:
##
##
```

	2.5%	25%	50%	75%	97.5%
alpha0	-2.737	-1.0073	-0.033040	0.9392	2.735
alpha1	-2.797	-0.9486	-0.006438	0.9323	2.745
beta0	-2.763	-0.9846	-0.039076	0.9057	2.829
beta1	-2.796	-1.0054	0.017855	0.9772	2.842
beta2	-2.835	-0.9629	-0.028170	0.9386	2.702
totalAbundance	6.000	46.0000	133.000000	393.2500	3465.175

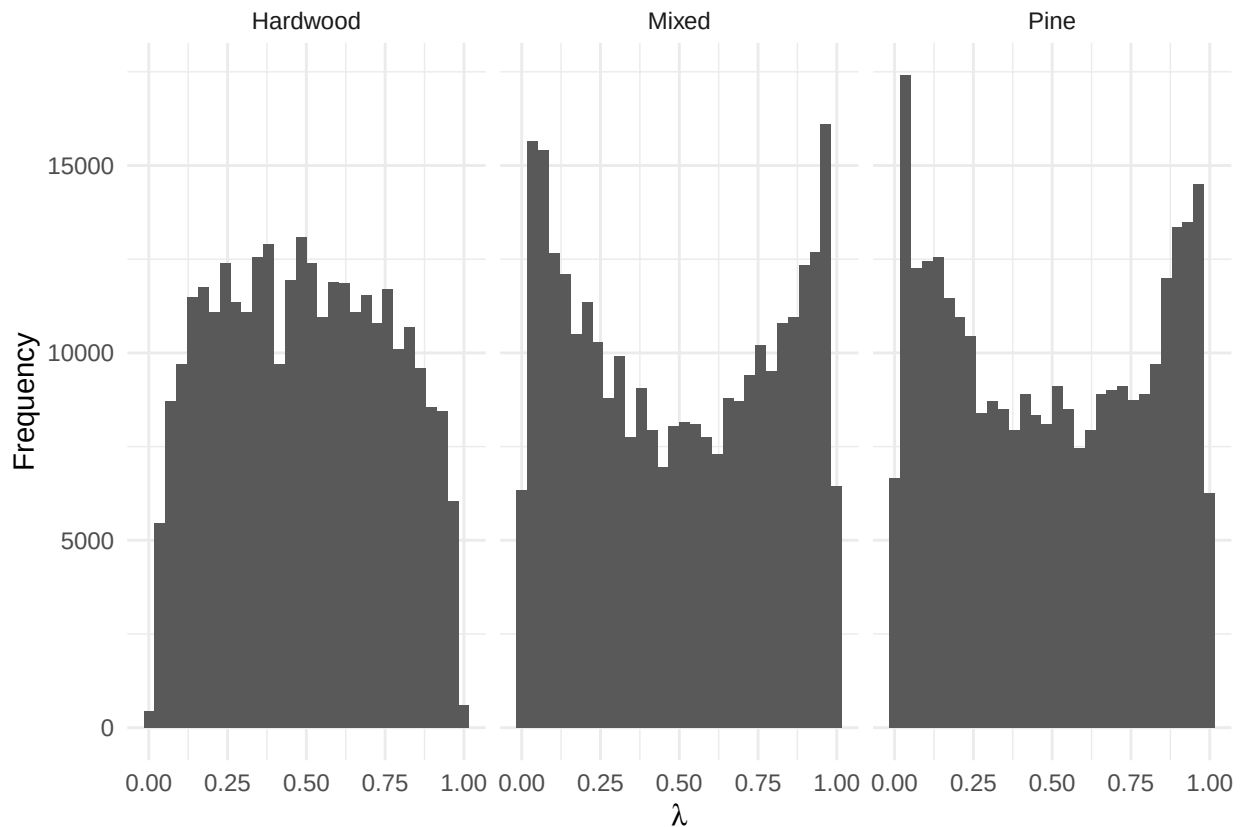
```
lambda.coef.prior <- as.matrix(jags.prior.samples[,c("beta0","beta1","beta2")])
lambda.prior.pred <- matrix(NA, nrow=n.iter, ncol=length(forest.pred))
```

```

for(i in 1:n.iter) {
  lambda.prior.pred[i,] <- plogis(lambda.coef.prior[i,"beta0"] + lambda.coef.prior[i,"beta1"]*forest.l)
}

# plot showing forest type on categorical horiz. axis and lambda on continuous vert. vertical axis
hardwood.prior <- as.vector(lambda.prior.pred[, 1:50])
mixed.prior <- as.vector(lambda.prior.pred[, 51:100])
pine.prior <- as.vector(lambda.prior.pred[, 101:150])
lambda.long.prior <- data.frame(
  lambda = c(hardwood.prior, mixed.prior, pine.prior),
  forest = rep(
    c("Hardwood", "Mixed", "Pine"),
    times = c(length(hardwood), length(mixed), length(pine))
  )
)
ggplot(lambda.long.prior, aes(x = lambda)) +
  geom_histogram(bins = 30) +
  facet_wrap(~forest, ncol = 3) +
  labs(
    x = expression(lambda),
    y = "Frequency"
  ) +
  theme_minimal()

```



```

lambda.long$distribution <- "Posterior"
lambda.long.prior$distribution <- "Prior"
lambda.combined <- rbind(lambda.long, lambda.long.prior)

ggplot(lambda.combined,
      aes(x = lambda, fill = distribution)) +
  geom_histogram(
    position = "identity",
    alpha = 0.5
  ) +
  facet_wrap(~ forest, ncol = 3) +
  labs(
    x = "lambda_hat",
    y = "Frequency",
    fill = "Distribution"
  )

```

'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.

